

TAFJORD, Norway

WMO: 012180

Lat: 62.2303N

Lon: 7.4217E

Elev: 17

StdP: 101.12

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.7	-6.8	-18.5	0.7	-3.9	-16.2	0.9	-3.4	12.3	2.8	11.8	3.0	1.9	180	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.5	23.7	15.6	22.0	15.0	20.5	14.4	17.4	21.3	16.4	20.0	15.6	18.8	3.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.8	11.3	19.0	15.0	10.6	17.9	14.1	10.0	17.0	49.0	21.3	46.1	20.0	43.6	18.9	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.7	8.4	DB	-9.8	27.1	1.8	2.8	-11.0	29.1	-12.1	30.7	-13.1	32.3	-14.4	34.3	(4)
(5)			WB	-10.7	18.7	1.8	1.1	-11.9	19.4	-13.0	20.0	-14.0	20.6	-15.3	21.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	1.5	1.3	3.0	6.8	10.1	12.8	15.4	14.9	11.8	7.9	4.6	2.2
	DBStd	6.17	3.90	4.02	3.46	3.45	3.37	2.69	2.76	2.74	3.28	4.18	4.29	4.39
	HDD10.0	1396	263	243	218	107	39	5	1	1	17	92	167	244
	HDD18.3	3887	521	476	477	346	254	167	98	113	197	325	412	502
	CDD10.0	567	0	0	1	11	43	90	169	151	71	26	6	1
	CDD18.3	16	0	0	0	0	0	2	8	5	1	1	0	0
	CDH23.3	70	0	0	0	0	2	7	38	22	1	1	0	0
	CDH26.7	8	0	0	0	0	0	0	6	2	0	0	0	0
Wind	WSAvg	2.9	4.3	3.9	3.7	2.9	2.6	1.9	1.6	1.7	2.1	3.1	3.7	3.5
Precipitation	PrecAvg	1102	124	109	110	53	48	64	67	75	96	109	113	134
	PrecMax	1530	355	224	462	193	110	170	108	167	244	251	310	278
	PrecMin	689	0	0	7	0	9	17	8	1	24	12	0	22
	PrecStd	209	93	63	98	39	23	34	26	36	60	56	88	65
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	12.4	13.3	18.5	22.8	24.2	27.0	26.1	22.9	20.8	16.9	14.2
		MCWB	6.0	6.3	6.7	9.9	13.4	15.3	17.5	15.7	14.3	12.8	10.0	7.9
	2%	DB	10.0	10.0	11.0	16.0	20.3	21.3	23.9	23.2	20.5	18.2	14.2	11.7
		MCWB	4.6	4.4	5.5	9.0	12.4	13.8	16.4	15.3	13.1	11.3	8.4	6.2
	5%	DB	8.6	8.4	9.5	14.3	18.1	19.4	21.8	21.5	18.8	15.8	12.4	10.0
		MCWB	4.0	3.4	4.5	8.1	11.3	13.0	15.9	15.1	12.4	9.3	6.9	4.8
	10%	DB	7.1	6.9	8.1	12.6	16.0	17.7	20.2	19.7	17.2	14.1	10.8	8.4
		MCWB	3.0	2.5	3.7	6.9	10.2	12.4	15.5	14.8	12.0	8.6	5.8	3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.3	6.7	7.5	10.9	14.9	16.2	18.8	18.3	15.8	13.5	10.6	8.2
		MCDB	10.7	11.4	11.6	17.3	20.7	22.0	23.1	21.6	19.3	19.2	15.8	12.9
	2%	WB	5.2	5.2	6.2	9.5	13.1	14.7	17.7	17.2	14.6	11.8	8.8	6.8
		MCDB	9.0	8.6	9.5	14.9	18.5	19.2	21.9	20.6	18.3	16.7	13.0	11.0
	5%	WB	4.2	4.0	5.3	8.5	11.9	13.7	16.8	16.4	13.6	10.5	7.6	5.6
		MCDB	7.7	7.0	8.6	13.2	17.2	17.7	20.7	19.5	16.7	14.5	11.5	8.8
	10%	WB	3.5	3.2	4.3	7.6	10.7	13.0	15.8	15.5	12.9	9.4	6.4	4.5
		MCDB	6.3	6.1	7.4	11.7	15.1	16.6	19.1	18.3	15.7	12.9	9.8	7.4
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.3	4.9	6.4	7.3	7.0	6.5	6.3	6.1	5.2	4.3	4.5
		MCDBR	6.1	6.4	6.9	8.5	10.7	10.4	9.8	9.6	8.6	7.6	6.6	6.9
		MCWBR	3.2	3.3	3.9	4.4	4.9	4.8	4.4	3.9	3.7	3.5	3.3	3.8
	5% WB	MCDBR	5.7	6.0	6.4	7.6	9.9	8.7	8.9	7.3	7.3	7.1	6.4	6.4
		MCWBR	3.3	3.4	4.0	4.3	5.1	4.4	4.6	3.8	3.8	3.8	3.6	3.8
Clear-Sky Solar Irradiance	taub	0.233	0.285	0.321	0.352	0.356	0.356	0.368	0.358	0.327	0.302	0.240	0.208	
	taud	2.044	2.173	2.155	2.156	2.222	2.326	2.380	2.409	2.470	2.358	2.094	1.954	
	Ebn at Noon	437	664	773	826	858	866	844	821	783	651	419	261	
	Edh at Noon	35	66	97	119	122	112	104	93	72	55	32	20	
All-Sky Solar Radiation	RadAvg	0.17	0.64	1.75	3.36	4.54	4.80	4.55	3.48	2.06	0.94	0.26	0.10	
	RadStd	0.04	0.15	0.31	0.43	0.40	0.48	0.40	0.35	0.29	0.15	0.06	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+1.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (38 neighbors)	+0.52	+0.70	N/A	N/A	N/A	N/A	-137	-186	+59	N/A

Nomenclature: See separate page